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Palaeosecular Variation at the Society Islands, French Polynesia

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Summary

The Society Islands are the peaks of volcanic cones which rise from oceanic depths in excess of 4 km and lie along a NW to SE lineament in the central Pacific basin. Oriented samples were collected from 53 sites on 5 of the younger islands of the chain. Geochronology indicates that all sites are younger than 3·4 My. Palaeomagnetic directions yield an angular standard deviation of 13·8° of the site VGP's about the geographic axis (which is indistinguishable from the mean VGP at the 95 per cent confidence level). This magnitude of palaeosecular variation is in accord with values predicted by secular variation models and suggests that the anomalous non-dipole field behaviour described at Hawaii has not extended to the region of French Polynesia.

1. Palaeosecular variation

Although the geomagnetic field is dipolar and axially symmetric when averaged over tens of thousands of years, at any given instant the position of the north magnetic pole measured from any other point on the Earth's surface may be offset from the rotational axis because of a combination of perturbing effects. These short time and small amplitude departures in direction (and intensity) of the geomagnetic field are termed secular variation, distinguished from the longer episodes of polar reversal and true polar wander. Whether or not these anomalies persist over very long periods of time is an extremely important question bearing on the nature and origin of the geomagnetic field. Another source of short term field perturbation is wobble of the dipole field about the rotational axis and its variation in intensity. These effects together contribute to the geomagnetic secular variation.

The variability of the geomagnetic field is commonly measured by either of two sets of data, one derived from the other. Palaeomagnetic field directions (I, inclination, and D, declination) may be determined directly from samples. Fluctuations in I and D are easily analysed by reference to the Fisher (1953) distribution and provide an estimate of the secular variation at a given locality over the time sampled. A more useful variable for palaeosecular variation analyses, however, is the virtual geomagnetic pole (VGP) (Cox & Doell 1960) which is simply that north magnetic pole which would produce the observed inclination and declination at a given latitude and longitude (assuming a perfect dipole field).

Virtual geomagnetic poles provide a convenient method for comparing secular variation, or angular dispersion, at widely separated localities. Scatter of the VGP's

can be analysed by Fisher's method or by a method proposed by Doell & Cox (1971), with very little difference. One great advantage in using VGP's is that angular dispersion resulting from wobble of the dipole field is constant with latitude whereas it is strongly latitude dependent when magnetic field directions are analysed. Significant differences in the estimated magnitude of secular variation arise between analyses employing VGP's and those using magnetic field directions so it is important to use the same type of data when comparing secular variation from separate localities and/or separate intervals of time.

The magnitude of secular variation as measured by the angular dispersion of palaeomagnetic data is latitude dependent, in accord with the latitude dependence of dipole field intensity. Models which attempt to explain this latitude dependence differ only in the relative contributions of a wobbling main dipole field of variable intensity and a westward drifting non-dipole field (Irving & Ward 1964; Creer, Irving & Nairn 1959; Cox 1962, Cox 1970). Combining the two effects, angular dispersion S_F can be written as

$$S_F = (S_D^2 + S_N^2)^{\frac{1}{2}}$$

where S_D is the angular dispersion of VGP's produced by the dipole wobble only and S_N is the angular dispersion due to the drifting non-dipole field. Reference to such a model is necessary when comparing magnitudes of secular variation from localities of differing latitude. Studies of palaeosecular variation have not been able to distinguish, at the 95 per cent confidence level (Cox 1969), one model from another, and in most cases model dispersions fall within the estimated confidence limits.

In a study of palaeosecular variation of Brunhes age ($<0{\cdot}7$ My) lavas from the island of Hawaii in the central Pacific basin, Doell & Cox (1965) found a much lower dispersion of VGP's than determined in similar studies of Brunhes age samples at other localities and lower than the model predicted dispersion for that latitude (20°). The present non-dipole field in the Hawaiian area is negligible and has been so over the few centuries of direct measurement of the geomagnetic field. From these observations and the low value of palaeosecular variation determined from Hawaiian lavas, Doell and Cox concluded that the central Pacific has been an area of persistently low non-dipole activity over the last 700 000 years. Because of this virtual absence of non-dipole perturbation, they believed the measured dispersion (11°) to be a good estimate of dipole wobble. To explain the attenuation of the non-dipole field in the central Pacific they suggested that a lateral inhomogeneity exists in the lower mantle beneath the region, which effectively shields or prevents generation of the non-dipole fluctuations. More recently Doell (1972 a, b, c) and Doell & Dalrymple (1973) proposed that low values of dispersion of VGP's from lavas at Oahu, Kauai, Niihau and Nihoa (all Hawaiian Islands) indicate that this central Pacific low in non-dipole field activity has persisted for at least the last 5 My. While similarly persistent low anomalies in the non-dipole field have not been discovered elsewhere, much importance has been attached to this result because of its potential power in indicating heterogeneities deep in the earth (Doell & Cox 1971).

The Society Islands of French Polynesia are recent volcanic features in the central Pacific basin, due south of the Hawaiian Islands and at about the same latitude, though in the Southern Hemisphere. The present study was undertaken to estimate the magnitude of recent secular variation in the French Polynesian area and to possibly establish the southern extent of the central Pacific low angular dispersion proposed by Doell & Cox.

2. Palaeomagnetic results from the Society Islands

The Society Islands (Fig. 1) are a group of volcanic peaks which stretch from NW to SE along a lineament roughly parallel to many other young island chains of the central Pacific basin (the Hawaiian Islands, the Marquesas Islands and the Austral

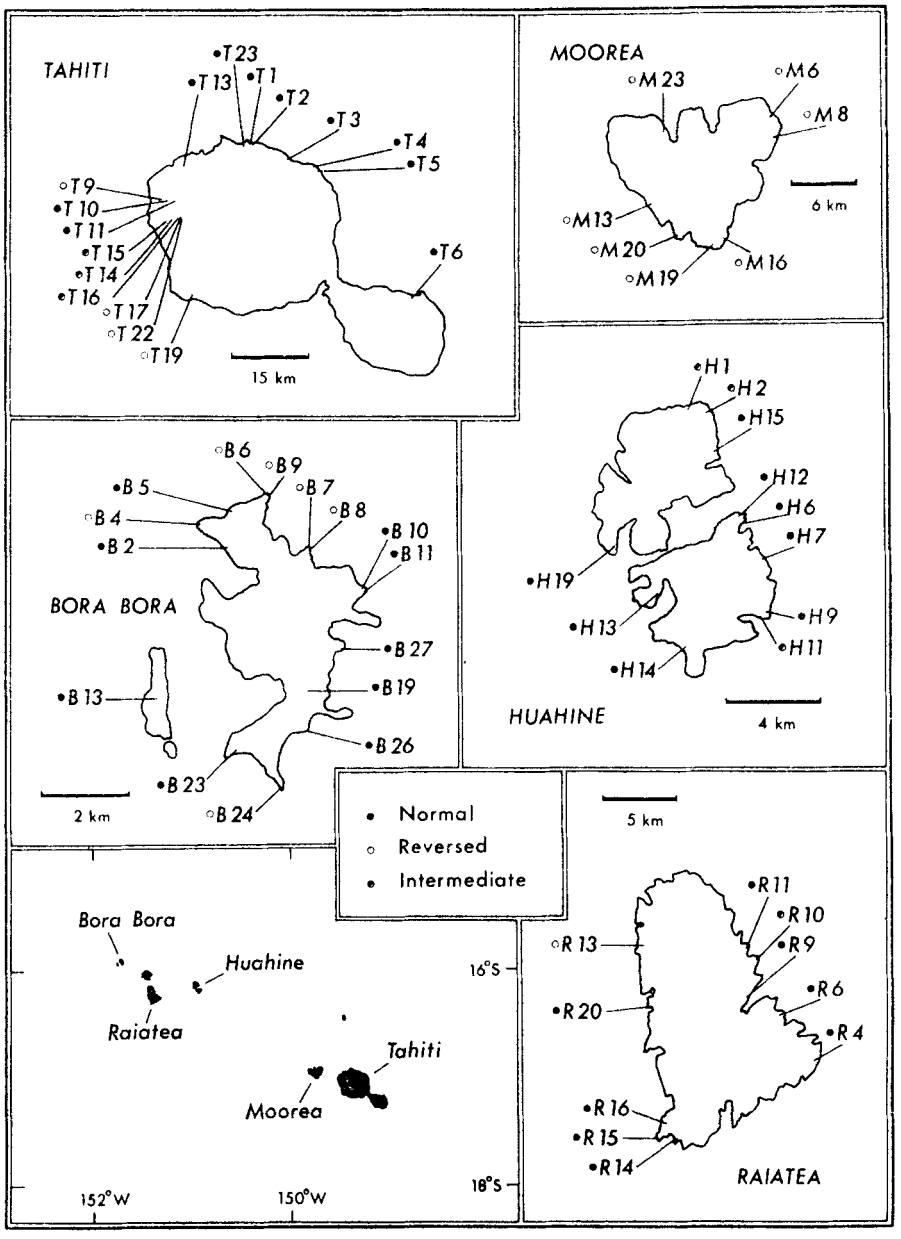


FIG. 1. The five sampled Society Islands and their location in the central Pacific basin, due south of Hawaii. Site localities are shown with determined polarities.

Islands, for example). In addition to this parallelism, these islands exhibit a migration of volcanism toward the SE end of from 10 to 12 cm/yr, in a sense opposite to the movement of the Pacific plate (Duncan & McDougall 1975). The islands have been constructed rapidly with very little overlap in time of activity at adjacent islands. The lavas of the various members of the Society chain belong to the alkali olivine basalt association of which the predominant representative is a slightly undersaturated basalt with subsidiary differentiates (hawaiites, mugearites and trachytes). Structurally each

Table 1

Site-mean palaeomagnetic results

Dyke or flow number	N	Peak cleaning field (mT)	Mean declination	Mean inclination	k	α_{95}	R	Polarity	VGP θ'	VGP ϕ'
Bora Bora (Mean site location is 16° 30′ S, 151° 44′ W)										
B2	4	20	1·7	−37·7	322·1	5·1	3·99	N	85·0	9·1
B4	4	20	166·0	39·6	79·7	10·3	3·96	R	−75·5	91·3
B5	4	40	347·9	−36·1	13·4	25·9	3·78	N	78·0	99·0
B6	5	20	194·0	42·6	67·0	9·4	4·94	R	−74·5	152·6
B7	4	NRM	164·3	32·7	531·7	3·9	3·99	R	−74·9	290·8
B8	4	20	182·4	26·3	753·1	3·3	3·99	R	−86·5	70·3
B9	4	20	183·1	16·5	58·1	12·1	3·95	R	−81·3	49·5
B10	4	10	359·5	−21·4	56·3	12·3	3·95	N	−84·6	203·1
B11	4	NRM	6·7	−35·7	392·4	4·6	3·99	N	82·7	326·3
B13	3	15	351·3	−53·4	13·5	34·9	2·85	N	70·8	50·5
B19	3	15	14·8	−37·9	88·6	13·1	2·98	N	75·1	319·2
B23	4	15	5·1	−28·8	230·7	6·0	3·98	N	84·9	286·0
B24	4	15	179·3	37·7	176·7	6·9	3·98	R	−85·2	215·7
B26	4	15	18·5	−37·6	143·7	7·6	3·98	N	71·8	315·6
B27	3	15	354·4	−37·3	544·1	5·2	2·99	N	78·6	103·4
Average	15		0·7	−35·1	44·1	5·8	14·68		87·0	15·1
Raiatea (Mean site location is 16° 50′ S, 151° 27′ W)										
R4	5	5	7·9	−40·1	452·3	3·6	4·99	N	80·2	339·2
R6	5	10	21·7	−17·1	323·1	4·2	4·98	N	67·3	280·8
R9	5	20	352·1	−20·0	353·1	4·0	4·98	N	80·0	157·2
R10	3	15	188·5	−29·7	13·5	34·8	2·85	I	−56·3	43·5
R11	5	15	7·7	−29·8	241·4	4·9	4·98	N	82·5	294·4
R13	4	15	179·0	37·1	1548·6	2·3	3·99	R	−85·8	220·3
R14	4	15	10·0	−39·3	122·2	8·3	3·97	N	79·0	330·8
R15	5	15	0·3	−36·8	260·1	4·7	4·98	N	86·1	23·4
R16	3	10	351·5	−29·3	243·6	7·9	2·99	N	81·8	124·3
R20	4	10	4·5	−31·6	329·1	5·0	3·99	N	85·6	304·9
Average	9		3·8	−31·5	48·1	7·4	8·83		86·2	305·1
Huahine (Mean site location is 16° 45′ S, 151° 00′ W)										
H1	5	15	319·0	−61·9	500·7	3·4	4·99	I	46·3	72·7
H2	4	15	344·1	36·9	52·1	12·8	3·94	I	49·7	186·1
H6	5	15	334·1	−27·5	344·0	4·1	4·99	N	65·0	120·7
H7	5	15	15·0	−44·5	21·5	16·8	4·81	N	73·0	336·0
H9	5	15	18·5	−29·4	19·4	17·7	4·79	N	72·1	298·9
H11	5	10	242·9	−21·1	155·3	6·1	4·97	I	−21·8	99·4
H12	4	10	350·6	−27·9	436·9	4·4	3·99	N	80·8	128·6

Table 1 (*continued*)

H13	5	15	337·3	−31·4	81·6	8·5	4·95	N	68·3	114·6
H14	5	15	8·2	−26·9	752·4	2·7	4·99	N	81·7	283·8
H15	5	10	33·5	−35·6	323·7	4·2	4·98	N	58·0	309·7
H19	4	10	29·4	−36·8	257·9	5·7	3·98	N	61·8	311·6
Average	8		2·2	−37·9	12·6	15·0	8·37		84·7	5·4
Moorea (Mean site location is 17° 32′ S, 149° 51′ W)										
M6	3	15	181·3	21·1	84·0	13·5	2·97	R	−83·3	220·0
M8	4	10	192·0	22·5	918·8	3·0	3·99	R	−76·9	95·5
M13	5	15	192·3	37·3	720·6	2·8	4·99	R	−77·8	138·2
M16	4	15	175·5	52·5	117·9	8·4	3·97	R	−73·8	223·8
M19	4	20	183·5	34·5	323·0	5·1	3·99	R	−86·3	144·7
M20	3	15	178·9	40·9	101·6	12·2	2·98	R	−83·9	219·0
M23	4	20	201·6	22·5	19·7	21·1	3·84	R	−53·7	247·9
Average	7		4·4	−34·9	38·9	10·8	5·87		85·4	323·4
Tahiti (Mean site location is 17° 40′ S, 149° 25′ W)										
T1	4	20	355·9	−34·9	467·8	4·2	3·99	N	85·7	95·6
T2	5	15	356·3	−13·2	50·4	10·8	4·92	N	78·6	192·0
T3	5	20	11·7	−32·4	379·4	3·9	4·99	N	78·8	302·9
T4	1	15	7·6	−25·8	—	—	—	N	81·6	273·7
T5	4	15	354·7	−31·2	48·5	13·3	3·94	N	84·9	126·5
T6	5	15	358·9	−28·6	97·9	7·7	4·96	N	87·3	188·7
T9	4	15	178·0	26·3	1380·0	2·4	3·99	R	−86·0	1·4
T10	5	NRM	338·2	−35·9	59·9	9·9	4·93	N	69·2	109·9
T11	3	15	1·5	−23·9	408·5	6·1	2·99	N	84·8	215·7
T13	4	NRM	1·0	−45·7	46·5	13·6	3·94	N	80·1	25·1
T14	4	15	332·6	40·4	32·9	16·2	3·91	I	41·4	176·0
T15	4	20	305·2	36·2	198·7	6·5	3·98	I	24·4	153·0
T16	3	20	196·3	72·3	90·1	13·0	2·98	I	−48·1	197·2
T17	4	20	178·2	64·6	20·1	20·9	3·85	R	−60·9	212·8
T19	3	15	169·2	5·7	545·3	5·2	2·99	R	−72·0	353·2
T22	5	NRM	186·5	35·9	96·0	7·8	4·96	R	−83·3	142·9
T23	3	NRM	357·3	−11·3	253·9	7·7	2·99	N	77·9	197·4
Average	14		358·6	−32·5	17·9	9·2	13·21		88·6	111·1
Average of normal poles	35		1·6	−33·4	27·2	4·6	33·71		87·9	338·1
Average of reversed poles	17		180·6	36·0	21·7	7·5	17·19		−86·9	198·9
Average of all sites	53		1·3	−34·3	25·4	3·9	51·92		87·7	355·2

N: number of samples per site α_{95}: radius of 95 per cent confidence circle *k*: precision parameter *R*: sum of *N* unit vectors

Table 2

Pacific palaeosecular variation

	N	S_T	S_W	$\bar{n}$	S_F	S_u	S_l	Ref.
Hawaiian Islands								
Recent	72	8·7	5	7·5	8·5	9·6	7·6	Doell & Cox (1965)
Brunhes only	130	13·5	5	7·0	13·4	14·7	12·4	Doell & Cox (1965) Coell (1972a)
Pre-Brunhes only	197	13·1	7	7·0	12·8	13·9	12·1	Doell (1972b, c), Doell & Dalrymple (1973)
All Hawaiian data	399	12·6	6	7·0	12·4	12·9	11·8	
Society Islands	53	14·6	9·6	4·0	13·8	15·9	12·2	This study
Easter Island	70	—	—	—	13·2	15·0	11·8	Isaacson *et al.* (1973)

N: number of sites
S_T: total angular dispersion
S_W: within-site angular dispersion
$\bar{n}$: average number of samples per site
S_F: between-site angular dispersion
S_u, S_l: upper and lower 95 per cent confidence limits on S_F, from Cox (1969)

island comprises one or more shield volcanoes which in their waning stages have collapsed to form central calderas. Eruptions of the differentiated series is the final phase of activity. Geochronology of Society Islands lavas (Duncan & McDougall 1975) has defined the age relationships within the group and has given estimates of the duration of activity at each island. Detailed geological descriptions are available elsewhere (Williams 1933; McBirney & Aoki 1967).

Previous palaeomagnetic work in two of the Society Islands by Tarling (1963) proved inconclusive. Samples from Tahiti and Bora Bora did not respond to magnetic cleaning and he concluded that a very intense and rapidly fluctuating non-dipole field was present in this region during the cooling of those lavas.

For the present study 60 dykes and flows at five of the Society Islands (Bora Bora, Raiatea, Huahine, Moorea and Tahiti) were sampled (see Fig. 1 for localities). Hand specimens were oriented in the field using a Sun-compass and later cored for laboratory measurement. Sites were selected to give the widest coverage possible, but confined to coastal cliffs and stream cut valleys. On average four samples per site were collected. The palaeomagnetic field directions, or natural remanent magnetizations (NRM's) of four cores from each oriented sample were measured using either a PAR fast spinner magnetometer or a Digico slow spinner magnetometer. NRM's were sufficiently scattered that alternating field demagnetization proved necessary. Sites were treated individually and varied in the strength of peak magnetic field needed to clean away secondary magnetic components. In general, however, 15 mT (150 Oe) AF was sufficient to produce stable directions. Results of the magnetic cleaning and final directions appear in Table 1. In addition Fisher statistics and virtual geomagnetic poles have been calculated for each site. Fig. 2 illustrates the distribution of VGP's of cleaned field directions for 53 normal and reversed sites. (Seven intermediate directions* are omitted in this discussion.) The average of 53 VGP's is indistinguishable from the rotational axis at the 95 per cent confidence level and uniform distribution of

* Wilson, Dagley & McCormack (1972) defined intermediate directions as those for which the VGP latitude $< 50°$. Their study employed a large number of Icelandic sites to show empirically that dipole moment (intensity) decreases dramatically for flows with VGP latitude $< 50°$. Others (Cox & Dalrymple 1967; Dagley *et al.* 1967) have arbitrarily chosen narrower bounds for 'intermediate' directions. Estimated angular dispersion is, however, extremely sensitive to inclusion of widely separated VGP's and in this study Wilson *et al.*'s more selective bounds have been used, leading to a minimum angular dispersion.

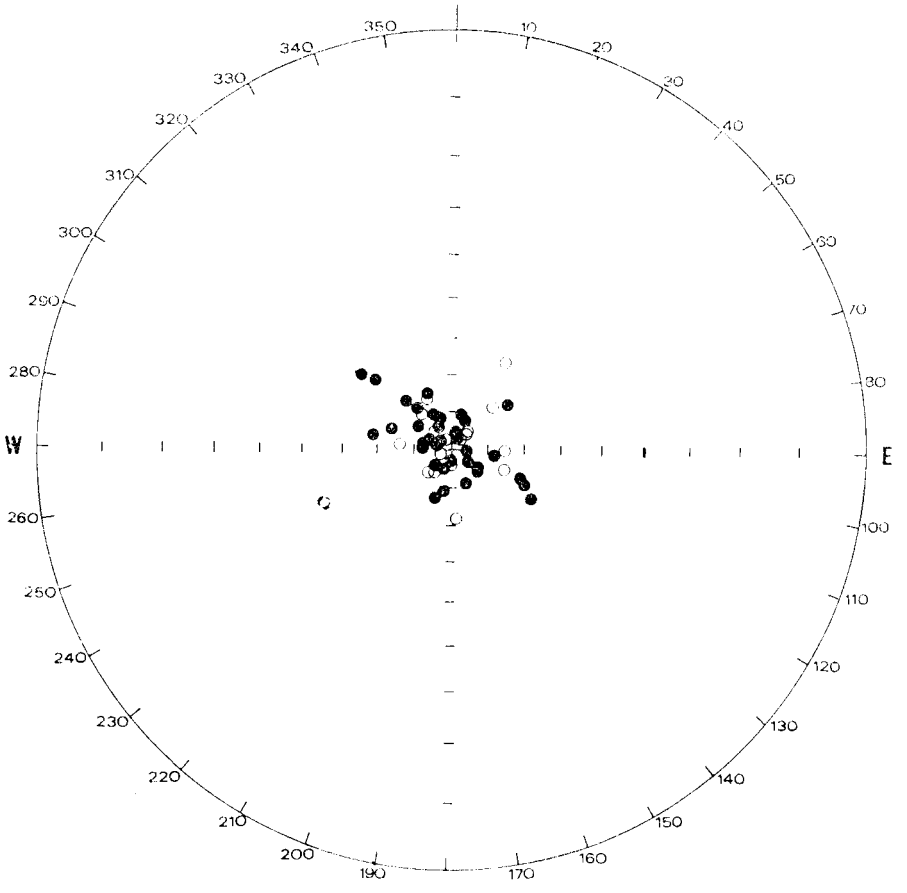


FIG. 2. Stereographic projection of site mean VGP's of remanent magnetization after magnetic cleaning (usually 15 mT or less). Solid circles indicate Northern Hemisphere VGP's (normal polarity epoch), open circles indicate Southern Hemisphere VGP's (reversed polarity epoch). The mean VGP is not distinguishable from the geographic axis at the 95 per cent confidence level.

the individual VGP's about the average insures that Fisher's method is appropriate for analysis of these data.

Geochronology (K–Ar) of volcanic material from these islands (Duncan & McDougall 1975) suggests that surface lavas and dykes at Bora Bora range in age from 3·1 to 3·4 My. At Raiatea ages vary from 2·3 to 2·5 My while at Hauhine from 2·0 to 2·3 My. Moorean ages lie between 1·4 and 1·6 My and at Tahiti the range is 0·4–1·3 My. Oriented samples used in the present study have been included in the geochronological investigation and determined ages all lie within the bounds cited. All ages and polarities are in accord with the polarity time scale. The time control provided by the geochronology indicates that the palaeomagnetic sites have sampled the geomagnetic field in the French Polynesian region at discrete points in time over the period 0·4–3·4 My.

Both normal and reversed sites were found at Bora Bora, Raiatea and Tahiti, while single polarities were found at Huahine (normal only) and Moorea (reversed only). No significant differences exist amongst mean VGP's (or dispersion about these means) for each of the five islands. For the purposes of this study the five data sets can

be combined to characterize the palaeosecular variation of the region over the time length of the geomagnetic field sampled by the lavas.

3. Dispersion and discussion

Several methods have been proposed for the estimation of angular dispersion of VGP's. Fisher's method (1953) determined the variance of the poles from their mean assuming each is an independent estimate of the position of the mean. Because there are two sources of dispersion, within site dispersion and between site dispersion, Watson & Irving (1957) proposed an analysis of variance technique which they termed a two-tier analysis of dispersion. Effectively, this method estimates the between site dispersion, which is a measure of the palaeosecular variation, and the within site dispersion, which is assumed to be constant for all sites. Doell & Cox (1971) remove within site dispersion in a similar estimate palaeosecular variation. All methods yield very similar estimates of angular dispersion of VGP's. An important question to be resolved is whether the angular dispersion at a given locality should be measured about the mean VGP or the rotational axis, if these are different. For young rocks which have experienced negligible translatitudinal tectonic movement, we expect the VGP's to average to the rotational axis if the geomagnetic field has been, on average, an axially centred dipole *and* a sufficiently long period of geomagnetic time has been recorded to sample the complete cycle of secular variation (both dipole wobble and non-dipole drift). Indeed, wide separation of the mean VGP from the rotational axis is evidence for insufficient sampling of the secular variation (Aziz-ur-Rahman & McDougall 1973). (The mean VGP for Society Islands sites is indistinguishable from the rotation axis at the 95 per cent level of confidence). The oldest sites sampled (Bora Bora) are younger than 3·5 My, implying maximum translatitudinal movement of 165 km, or 1·5 degrees (Duncan & McDougall 1975). For younger sites, the movement has been correspondingly less. Hence a small correction may be applied to the position of the rotational axis about which dispersion is measured, for each island. For such a calculation, this ' tectonic correction ' made little difference so in this study angular dispersion of VGP's has been measured about the present rotational axis. For comparative purposes, all studies from the Hawaiian Islands have been recalculated to this form if necessary.

Table 2 presents the results of palaeosecular variation calculations for the 53 Society Islands sites of this study as well as data from other studies in the Pacific region. S_F, the angular standard deviation of site VGP's, has been calculated by the Doell & Cox method, with respect to the rotational axis. S_u and S_l are the upper and lower 95 per cent confidence limits on S, determined from Cox (1969), and N is the number of site VGP's employed in each calculation. Fig. 3 presents these results graphically.

An angular standard deviation of VGP's about the rotational axis of 13·8° characterises palaeosecular variation in the region of the Society Islands for the period 0·4–3·4 My. This estimate is well constrained by the 95 per cent confidence limits, S_u, S_l due to the size of the sample, 53. The predicted dispersion, from reference to secular variation models*, lies within this range so that non-dipole field activity during the time interval recorded has not been anomalous with regard to other localities. Palaeosecular variation recorded in Brunhes age lavas at Easter Island (27·1° S 109·4°W) is likewise in accord with model dispersion (Isaacson *et. al.* 1973), so that the southern extent of the Pacific non-dipole field anomaly recorded at Hawaii lies north of the Society Islands and Easter Island.

Combining all Hawaiian data, 399 site VGP's yield an angular standard deviation of 12·4° with very narrow limits. A *t*-test indicates that palaeosecular variation has

* The model angular standard deviation at latitude 20° equals 14·2° (Cox 1970).

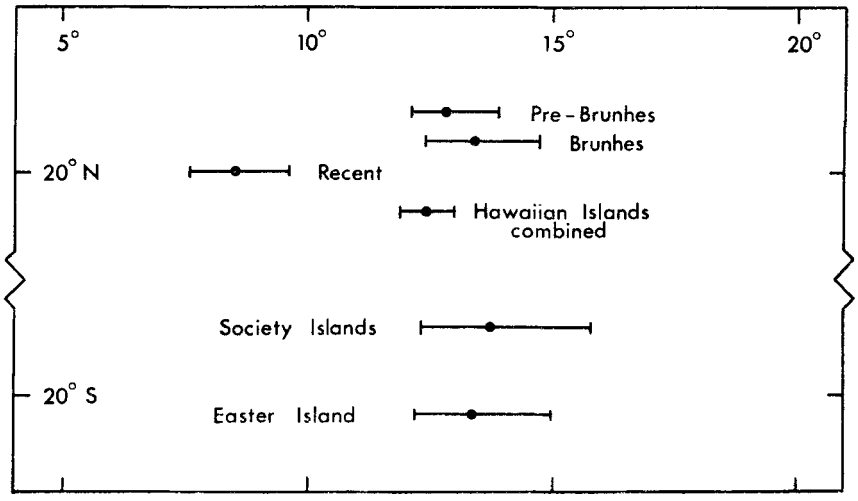


FIG. 3. Graphical comparison of results from palaeosecular variation studies on lavas from volcanic islands in The Central Pacific basin. Angular dispersion, measured about the geographic axis, is expressed in degrees along the abscissa. Error bars are the upper and lower 95 per cent confidence bounds from Cox (1969). The Hawaiian data have been divided chronologically, as explained in the text. Table 2 presents these results numerically.

been significantly lower in the Hawaiian area over the last 4 My than in the French Polynesian and Easter Island areas. Closer inspection reveals that the angular dispersions resulting from various Hawaiian studies (Doell & Cox 1965; Doell 1972 a, b, c; Doell & Dalrymple 1973), each calculated about the rotational pole rather than the mean VGP, fall into two groups. The groups are best separated chronologically. VGP's for sites from Recent eruptions (Kau and Puna series on Hawaii) show a very low angular dispersion, 8·5°. Indeed, this palaeosecular variation is lower than the 11° attributed to dipole wobble alone (Doell & Cox 1971), suggesting a considerably smaller component from that source of dispersion. Alternatively, as suggested by Aziz-ur-Rahman & McDougall (1973), these Recent eruptions have not completely sampled the cycle of secular variation. Excluding the Kau and Puna series, Brunhes age site VGP's are more dispersed, with an angular standard deviation of 13·4°, indistinguishable from older Pre-Brunhes Hawaiian palaeosecular variation, 12·8°. Those studies of Recent lavas from the Hawaiian area, then, show significantly lower angular dispersion of VGP's and suggest a fundamentally lower level of non-dipole field activity during the last several thousand years. The magnitude of palaeosecular variation in the Hawaiian region from sites older than the Recent Kau and Puna eruptions is comparable to that calculated from the Society Islands and at Easter Island sites and not anomalous when referred to model dispersions based on globally distributed studies. And, while the Central Pacific Basin is presently an area of subdued non-dipole field activity, the anomaly has not persisted for times older than Recent nor is there evidence for its extension south to the French Polynesian region for any time over the last 4 My.

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